Auditing Reference Distributions for Observation-Replacement Return Spans

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Abstract

An observation-replacement return span measures how a specified input intervention changes a policy's return. Interpreting that span requires choices about the reference population, replacement distribution and evaluation process. We audit these choices in a binary-contract simulator, controlled synthetic populations and six control-task configurations. We retain training-seed variation, raw endpoint returns, failed competence gates and unresolved references. In the historical market example, five policies have mean return -0.770 dollars per episode with a crossed seed/day interval $[-1.509, 0.011]$. Reference comparisons remain descriptive because market policies are not exchangeable with the synthetic bank. Conditional synthetic calibration and the control-task results are reported with their denominators. The zero-strength population still has positive full-input replacement spans, illustrating that return sensitivity can arise without settlement-predictive inputs. An exact finite-pool example shows why the legacy reference-centered rank/normal rule lacks a general exchangeability guarantee; standard symmetric ranks address that separate statistical requirement. The audit supports careful interpretation of particular interventions, without inferring feature-level fidelity, absent information or explanation meaning from a single return-span comparison.

1 Question and scope

An explanation can describe a policy's decisions, its predictions, or the consequences of its actions. These are different targets, including when the policy performs poorly. SVERL makes this distinction explicit and assigns feature contributions to specified characteristic functions (Beechey et al., 2023, 2025). A faithful explanation of a costly decision can be useful for debugging it. Low return alone does not make the explanation empty.

Our question concerns a narrower diagnostic: how should an observed observation-replacement return span be compared with spans from a bank of reference policies? We examine the implemented comparison, the financial state retained by an intervention, and the evidence needed to interpret reference widths and classifications. The contributions are a reproducible computational audit, explicit examples of interpretation failures, and empirical comparisons under declared training budgets. We do not propose a universal null for reinforcement learning or a new general rank-test theorem.

The study is retrospective. Its configuration was fixed for the selected rerun after exploratory development on this historical data. Repeating that window does not create a prospective holdout. All numerical tables are generated from the selected release; the training source is `dcc4e3f31c69`. Reduced-budget execution runs cannot supply these tables.

2 What the endpoint statistic identifies

Let π be a fixed policy, M an environment, D an evaluation population, and I a specified replacement intervention. For a feature set S, let $v_{\pi,M,D,I}(S)$ denote expected episode return when the intervention retains the coordinates in S. The endpoint span is

$$\Delta(\pi, M, D, I) = v(N) - v(\emptyset). \tag{1}$$

We estimate both endpoints by replaying the same held-out episodes. The full endpoint supplies original policy inputs; the empty endpoint applies the declared replacement at every decision. The intervention also changes the policy's actions and subsequent inventory. Therefore this is a property of the policy, environment, intervention and evaluation distribution together.

For an efficient Shapley decomposition, $\sum_i \phi_i = v(N) - v(\emptyset)$ (Lundberg and Lee, 2017). Reading the endpoints avoids estimating individual coordinates merely to recover their sum. It does not validate that decomposition: allocations with the same sum receive the same diagnostic regardless of how they assign feature credit. Estimation accuracy for a ranking is another question; our implementation's gap diagnostic does not implement the formal ranking procedures of Goldwasser and Hooker (2024).

The intervention is also distinct from SVERL's conditional policy-occupancy construction. The extended framework defines outcome explanations through partial-information behaviour at the state being ex-

plained, with the original policy elsewhere (Beechey et al., 2025). Our empirical flat-inventory background and repeated replacement at all steps differ in distribution and locality. The experiments do not test or refute that theoretical construction.

2.1 Dependence, fidelity and task utility

Parameter-randomization sensitivity asks whether an explanation depends on learned parameters (Adebayo et al., 2018). It does not, by itself, establish feature-level fidelity or useful task performance. Limitations of such checks and refinements to their evaluation are established topics (Binder et al., 2022; Hedström et al., 2024). A bank of independently initialized policy spans is a different experiment from progressively reinitializing the layers of one trained model.

Integrated gradients satisfy completeness for their chosen model output (Sundararajan et al., 2017). Our auxiliary IG implementation explains a chosen-action probability relative to a zero normalized input. That probability difference is not the return intervention in Equation 1. We therefore do not claim cross-family return-test agreement from these auxiliary profiles.

3 Reference diagnostics and rank validity

The legacy rule compares an observed span T with B reference spans $T_1, \dots, T_B$. It combines a reference-centered empirical tail score

$$p_{\text{legacy}} = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{|T_b - \bar{T}_{\text{ref}}| \geq |T - \bar{T}_{\text{ref}}|\}}{B+1} \tag{2}$$

with a two-sided normal approximation to $z = (T - \bar{T}_{\text{ref}})/s_{\text{ref}}$. Both criteria must meet the declared level $\alpha = 0.05$. We retain this rule as a diagnostic of the original workflow, not as an assumption-free test. Insufficient rank resolution or $s_{\text{ref}} \leq 10^{-12}$ makes its comparison unresolved. This convention does not mean that every test against a point-mass reference is invalid.

Proposition 1. *The legacy combined rule does not have a general finite-sample exchangeability guarantee.*

Proof. Uniformly designate one observation in a fixed pool containing $-1, +1$ and 18 zeros; the remaining 19 entries are references. The resulting observation and references are jointly exchangeable. Either extreme receives legacy tail score $1/20 = 0.05$ and $|z| = 20/\sqrt{19} > 4.58$, so both meet the normal criterion. The rejection probability is $2/20 = 0.10$. This is a finite-pool example, not an error estimate for the independent synthetic benchmark. □

For comparison, let $T_0 = T$ and compute the pooled center $\bar{T}_{\text{pool}} = (B+1)^{-1} \sum_{b=0}^{B} T_b$. The symmetric two-sided rank uses scores $A_b = |T_b - \bar{T}_{\text{pool}}|$ and

$$p_{\text{sym}} = \frac{\sum_{b=0}^{B} \mathbf{1}\{A_b \geq A_0\}}{B+1}. \tag{3}$$

We also report the upper-tail raw-span rank. Both include the observation and count ties conservatively. Under joint exchangeability and permutation-equivariant scoring, their ranks are super-uniform by the established finite-rank argument (Kuchibhotla, 2020; Hemerik and Goeman, 2018). Conditional on an unordered score multiset, at most an α fraction of indices can have an inclusive upper-tail rank at most α; exchangeability makes the observed index uniform. The symmetric rank assigns each extreme in the example $2/20 = 0.10$.

That guarantee is marginal over the bank and the observed draw. It does not bound error conditional on each realized reference bank. It also does not supply exchangeability between a historical market agent and policies trained on synthetic populations or corrupted inputs. Connections between attribution and conditional-independence testing require explicit modeling assumptions, as in SHAP-XRT (Teneggi et al., 2023); the return setting does not inherit those guarantees automatically.

4 Implemented experiments

4.1 Market accounting and evaluation

The pinned corpus contains 6,425 binary-contract episodes with 14 decision rows each. Chronological training, validation and test counts are 3,854, 1,283 and 1,284, with four boundary episodes purged. The 18 inputs comprise 14 market/spot coordinates normalized using training statistics and four active-inventory coordinates. Actions target $-100, 0, +100$ signed yes-equivalent contracts. A reversal can trade 200.

The full-input replacement background contains 256 rows sampled from all evaluation steps while inventory is held flat. Each intervened episode draws a fresh whole background row at every decision. This replaces the clock as well as market and inventory coordinates; it is an empirical intervention, without a guarantee of compatibility with the recipient's state.

The assumed execution model buys at the supplied ask and sells at the bid. It charges the per-order fee $\text{ceil}_{\text{cents}}[0.07|C|p(1-p)]$, with zero impact by default, full fills and no funding or queue simulation. Candle-derived quotes are coarse proxies; these assumptions do not establish exact historical fees or live executable performance. The default terminal mark is the recorded binary settlement, with no additional settlement fee in the model. Equity increments telescope to final profit and loss at $\gamma = 1$.

Five market seeds train for 100 PPO updates under each cost objective. Their greedy policies are eval-

uated on every held-out episode. The critic predicts stochastic policy return in dollars, not settlement probability; its auxiliary calibration uses eight policy rollouts per episode. Both rollout implementations agree on every acted-on observation and greedy action for all ten checkpoints, with maximum accounting discrepancy $\leq 3 \times 10^{-14}$. Primary uncertainty crosses training-seed resampling with shared UTC-day blocks. There are only 14 test-day blocks, limiting inference about temporal variation.

4.2 Synthetic populations and reference banks

The balanced synthetic population has independent binary settlements, constant price 0.5, spread 0.02, and noisy features independent of settlement. Flat earns zero. The planted population adds a noisy settlement signal, repeated across the 14 steps. Its first-observation sign-and-hold strategy is a reference policy, not a proven optimum when aggregation, delayed entry and abstention are possible.

The primary bank has 24 independently seeded policies, each trained for 40 updates on 1,200 synthetic episodes and evaluated on an independent draw of 1,200 episodes with its own training normalizer. A 99-draw sensitivity retains the same procedure. The conditional calibration uses 30 independent observed trials at each strength $0, 0.2, 1$, reusing the primary bank. Each trial retains both endpoint returns. We report pointwise Wilson intervals conditional on that bank, alongside unresolved counts and both standard rank comparators. These rates include the specified learning procedure and budget. They do not isolate test sensitivity from learning success or quantify bank-to-bank error.

4.3 Control tasks and competence gates

We evaluate CartPole, Acrobot, MountainCar, Pendulum with nine discrete torques, deterministic 4×4 FrozenLake, and CliffWalking capped at 200 steps. The control trainer is Stable-Baselines3 PPO (Raffin et al., 2021), with separate two-layer 64-unit actor/value trunks and four environment copies. Observed agents request 400,000 interactions; each blinded reference requests 30,000. Complete rollout batches can exceed those requests. Each construction has 24 reference draws, and endpoint evaluation uses 20 episodes. Five independently trained observed seeds are separate from correlated training checkpoints.

The parameter reference uses fresh orthogonally initialized actors, with no learning. The environment reference trains agents with independent coordinate-wise Gaussian observations whose moments are estimated from a random-policy rollout, leaving transitions and rewards intact. Both are evaluated on the original task with full inputs and with repeated background replacement; the background contains 512 random-policy observation rows. Gaussian blinding can violate within-state constraints. A separate whole-row resampling sensitivity preserves donor row support, without guaranteeing consistency with the recipient trajectory. Initialization and blinded learning are different reference procedures, without an established exchangeability relationship to competent observed agents.

Declared competence thresholds are respectively 475, −100, −110, −300, 0.8 and −20, measured on 25 episodes, together with an absolute measured endpoint span above 10^{-6}. A failed gate does not establish low detection power. Neither reference-width ratios nor standardized distances are power multipliers. All failed and unresolved cases remain in the results.

Training objective	Seeds	Mean	Seed SD
Fees and spread	5	−0.770	0.354
Zero cost	5	0.056	0.984

Table 1: Held-out market return in dollars per episode. Seed spread is the sample SD across seed means, not episode SD or a standard error. Cost and zero-cost agents were trained separately. The primary crossed seed/day interval is $[-1.509, 0.011]$. Non-rejection would not establish equivalence.

Case	Span	Ref. SD	z	Decision
Planted signal	75.627	3.857	17.61	outside
Zero strength	12.436	3.857	1.23	within
Market seed 0	8.848	3.857	0.30	within

Table 2: Primary synthetic-bank diagnostics. "Outside" denotes the legacy two-part classification, not a finding of explanation fidelity. Market comparisons are descriptive. Undefined distances are shown explicitly.

5 Empirical results

5.1 Historical market and primary references

The market policies have mean return −0.770 dollars per episode (Table 1). The primary centered-bootstrap probability is 0.0467. This approximate probability and the percentile interval are different summaries and need not be dual near the decision level. We report both, without selecting one to support a stronger claim. Neither a negative mean nor a comparison with flat proves absence of predictive information or makes a policy's explanation meaningless.

Table 2 compares the fixed primary cases against the declared bank. All five market-seed comparisons and the larger-bank sensitivity are retained in the release. Across 5 market seeds, the primary bank gives 0 outside,

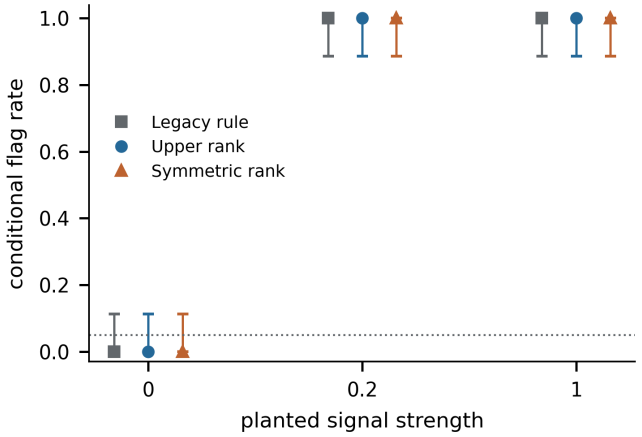


Figure 1: Conditional trial rates with pointwise intervals for one reused reference bank. A standard rank guarantee under exchangeability is marginal over banks; these intervals do not establish such a guarantee on the market.

Task	Return	Elig.	Env. O/U	Param. O/U
CartPole	500.0	5/5	5/0	1/0
Acrobot	−75.8	5/5	5/0	5/0
MountainCar	−185.4	0/5	0/5	0/5
Pendulum	−166.9	5/5	5/0	5/0
FrozenLake	1.0	5/5	5/0	5/0
CliffWalking	−200.0	0/5	0/5	0/0

Table 3: Independent control seeds. Eligibility requires both the declared return gate and measured endpoint dependence. Classification counts retain all seeds and report unresolved references separately. Each task uses its native reward units. O/U denotes outside/unresolved counts out of five seeds.

5 within and 0 unresolved classifications. Their spans range from 7.409 to 11.846. The larger bank gives 0 outside, 5 within and 0 unresolved classifications across the same 5 market seeds. Changing a bank can change its center, width and classification while also changing training or evaluation distributions. Those differences do not identify a unique information-removal mechanism.

5.2 Conditional calibration and controls

At zero strength, the legacy rule flags 0/30 resolved trials (conditional interval [0.000, 0.114]), with 0 unresolved trials. The upper-tail and symmetric ranks flag 0/30 and 0/30, respectively. Their conditional intervals are [0.000, 0.114] and [0.000, 0.114]. With 30 trials, even zero flags leave a substantial upper uncertainty bound. One positive-control success is therefore not a power estimate. The signal strength sweep also changes what the learning procedure can acquire within its fixed budget.

Of 30 independently trained control agents, 20 meet both eligibility gates. Of those, 20 have two resolved

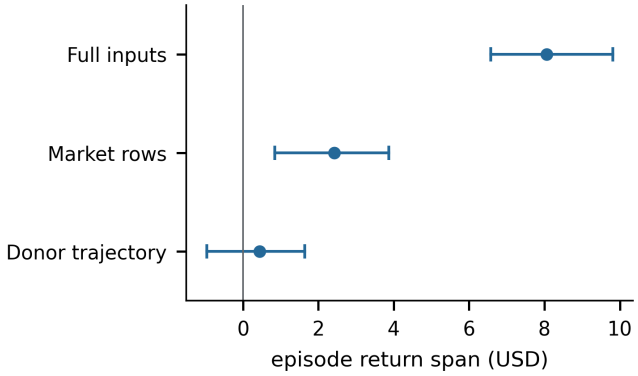


Figure 2: Market return spans under three specified interventions. Intervals cross training seeds and common UTC-day blocks, conditional on fixed masking draws. Differences do not isolate inventory as their sole cause.

references; their classifications differ in 4 cases. No unresolved comparison is counted as reference agreement. This table supports statements about these configurations and reference banks. It does not establish a universal ranking of reference constructions. A reference can collapse when its policies have identical returns, and the legacy normal-scale rule then cannot decide. That outcome differs from a resolved non-rejection and from a trained observed agent failing to learn.

5.3 State preservation and attribution suppression

The market state controls use the five saved cost-enabled checkpoints. Each synthetic case trains five new policies for 60 updates, using 1,200 episodes per training draw and 1,200 common independent evaluation episodes. Full replacement changes inventory inputs as well as market inputs. The market-row and donor-trajectory controls retain the real clock and all four inventory coordinates of the intervened rollout. A donor row preserves empirical row support, and a donor trajectory preserves its own market sequence. Neither guarantees joint consistency with the recipient's financial state or reward.

The full-input replacement span is 8.065, with interval [6.579, 9.816]; the donor-trajectory span retaining inventory and clock is 0.443, with interval [−0.963, 1.634]. These runs fix masking draws within each analysis; their recorded seeds differ from the primary comparison's draws for some policies. The zero-strength population also has a full-input span of 9.859, with interval [4.075, 16.183]; its donor-trajectory span is −0.118, with interval [−0.795, 0.362]. Positive sensitivity in this generator need not indicate settlement-predictive information: replacement can change trading and its costs. These controls do not isolate one cause of the span differences.

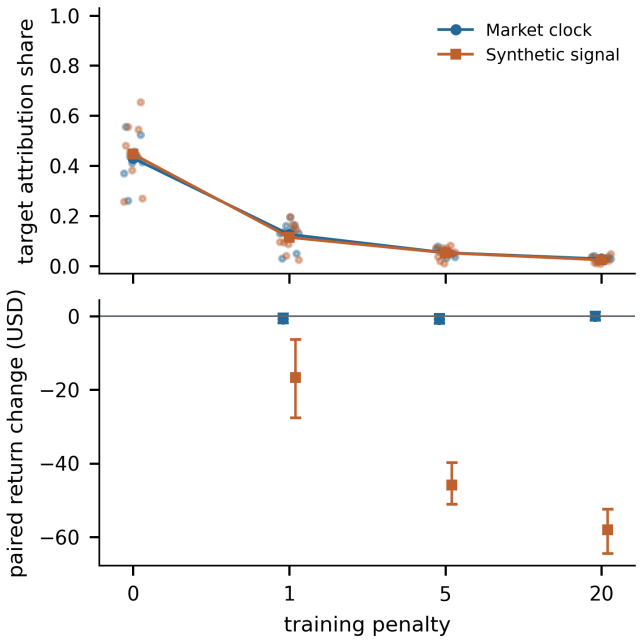


Figure 3: All declared penalties. Upper panel shows seed-specific target shares and their means; lower panel shows paired return changes with crossed seed/block intervals. The two populations target different features.

Retaining inventory also retains some price information. If $z = q/100$ is scaled nonzero position, e its entry price and $u = q(P-e)/100$ scaled unrealized profit, then

$$P = e + u/z. \tag{4}$$

The identity holds up to feature precision. Thus a financially consistent control retaining these coordinates cannot also hide all current-price information when $z \neq 0$. Conditional information beyond retained inventory can still be a legitimate target; this identity is not an impossibility result for conditional explanations.

The steering experiment penalizes input gradients during training and measures the resulting behaviour-attribution share and held-out return separately. An auxiliary single-seed comparison examines greedy behaviour; it cannot establish behavioural equivalence across the full experiment. The penalty targets the expiry clock in the market and the planted signal in the synthetic case, with ten seeds and penalty 20 fixed for the reported comparison. Each steering policy trains for 60 updates. Synthetic training and independent evaluation each use 1,500 episodes; market evaluation uses the complete test partition. All penalties are retained. For the market clock, the mean target share changes from 43.0% to 2.9%; the paired return change is −0.072 dollars, with interval [−1.072, 1.066]. For the synthetic signal, the mean target share changes from 44.7% to 2.4%; the paired return change is −58.098 dollars, with interval [−64.563, −52.458]. A non-significant return contrast is not equivalence or noninferiority, and changed task return alone cannot establish that the resulting policies are behaviourally identical.

6 Why settlement shuffling changes the question

Global settlement permutation preserves feature arrays and the marginal settlement rate. It changes their relationship to rewards. This matters even in a calibrated one-period example with no incremental price-measurable edge. Suppose $Y \in \{0,1\}$, $P \in [0,1]$ and $\mathbb{E}[Y \mid P] = P$. Every bounded price-measurable position $q(P)$ has frictionless gross expected payoff

$$\mathbb{E}[q(P)(Y-P)] = 0. \tag{5}$$

If Y^* is replaced by an independent draw from its marginal settlement law, with mean $\mu = \mathbb{E}[P]$, choosing $q^*(P) = \operatorname{sign}(\mu - P)$ instead gives

$$\mathbb{E}[q^*(P)(Y^*-P)] = \mathbb{E}|\mu - P|, \tag{6}$$

which is positive for a nonconstant price. Fees and execution constraints can reduce this gross payoff. The example does not establish live profit or efficiency of the historical corpus. It shows that independence of settlement and price is a different hypothesis from calibrated prices with no additional edge. Price remains reward-relevant through acquisition cost even after losing settlement-predictive information.

Within-price-bucket label changes and correlations are descriptive diagnostics, not estimates of conditional information removal. A conditional null may legitimately retain predictive information in its conditioning price. Failure to meet a finite set of heuristic bucket thresholds does not prove conditional randomization impossible or vacuous. A sequential test would need a justified joint conditional law for paths and outcomes.

7 Limitations and next experiments

The financial model uses assumed costs and coarse quotes, the market period is fixed, and seed/day inference rests on few time blocks. Synthetic calibration is conditional on one bank and has limited precision at rare error rates. Control competence is configuration-specific; failed tasks remain evidence about training rather than about detection sensitivity. No reference width alone identifies an explanatory mechanism.

This audit distinguishes computational correctness from scientific identification. Complete replay parity, corrected KernelSHAP coalition-size mass and exact-game tests strengthen the implementation evidence. They do not turn the endpoint sum into a feature-fidelity test. Auxiliary ranking, IG and parameter-randomization diagnostics address different targets and

assumptions. The same applies to secondary objective and reference sensitivities: some masked spans retain means without per-episode masked arrays, limiting independent endpoint reconstruction.

The next useful experiments are repeated independent reference banks with more null trials, competence-controlled comparisons with matched learning budgets, and conditional path randomization in a simulator with a known joint law. A separately frozen temporal market holdout would test external robustness. These additions would strengthen calibration and generalization; they are not prerequisites for honestly reporting the present descriptive audit.

Reproducibility. The repository pins the corpus and splits, retains raw returns and model hashes, records numerical runtime and package versions, and verifies source against the training commit. The full runner and release verifier are separate from execution-only smoke checks. Manuscript tables, figures and summaries share one derived statistics artifact; the three document profiles share this source. No claim of compliance with a particular venue's format is made.

A Additional numerical records

Seed	Span	z	Decision
0	8.848	0.30	within
1	7.409	−0.08	within
2	11.846	1.08	within
3	8.003	0.08	within
4	8.634	0.24	within

Table 4: Every market-seed comparison against the primary bank. These remain descriptive comparisons with a synthetic reference population.

Policy	Mean return
always-flat	0.000
buy-and-hold	−1.608
random	−18.532
mean-reversion	−11.615
momentum-fallback	−16.091
logistic-refit	−5.729

Table 5: Descriptive cost-enabled baseline returns on the same held-out episodes. A lower mean is not a statement about which model has a faithful explanation. The momentum fallback is labeled explicitly.

Stored auxiliary analyses. The release also retains the reference-training budget sweep, the larger-bank sensitivity, nearest-neighbor masking, explanatory target proxies, layer randomization and horizon-width studies. Nearest-neighbor replacements do not provide a known conditional law. Horizon lengths change clock inputs and trading opportunities together; their log-log fits are descriptive. Unresolved comparisons and undefined correlations are retained rather than converted to agreement, no signal or equivalence.

Population	λ	Share	Return	Change CI
Market	0	43.0%	−1.59	baseline
Market	1	12.6%	−2.30	[−1.81, 0.64]
Market	5	5.3%	−2.40	[−1.90, 0.47]
Market	20	2.9%	−1.66	[−1.07, 1.07]
Signal	0	44.7%	45.46	baseline
Signal	1	11.4%	28.74	[−27.70, −6.38]
Signal	5	5.3%	−0.43	[−51.12, −39.83]
Signal	20	2.4%	−12.64	[−64.56, −52.46]

Table 6: All steering penalties. Attribution shares concern the specified behaviour proxy. Return differences use common evaluation blocks and paired training seeds. An interval containing zero does not establish equivalence.

Case	Intervention	Span	95% CI
market	Full inputs	8.06	[6.58, 9.82]
market	Market rows	2.42	[0.84, 3.87]
market	Donor trajectory	0.44	[−0.96, 1.63]
null	Full inputs	9.86	[4.07, 16.18]
null	Market rows	−0.16	[−0.88, 0.28]
null	Donor trajectory	−0.12	[−0.79, 0.36]
signal	Full inputs	73.78	[70.55, 76.81]
signal	Market rows	63.12	[61.11, 65.05]
signal	Donor trajectory	47.38	[45.16, 49.60]

Table 7: State-preservation spans and crossed seed/block intervals.

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